

1.7 Differentiation

Pure Mathematics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Derivative is instantaneous gradient and rate of change. Power rule $d(x^n)/dx = nx^{n-1}$; use product, quotient and chain rules where needed. Stationary points satisfy derivative zero, but classify with second derivative or sign change. Related rates require units and sign.

Worked example: follow the reasoning

For $y = x^3 - 3x$, $y' = 3x^2 - 3$; stationary $x = \pm 1$, and $y'' = 6x$ classifies $x = -1$ maximum and $x = 1$ minimum.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Differentiate $y = (2x+1)^3$ and find gradient at $x=1$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Chain rule $y' = 3(2x+1)^2 \times 2 = 6(2x+1)^2$ [2]; at $x=1$ gradient 54 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.